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Round Robin CPU Scheduling Algorithm

Completed

Write a C program to implement the Round Robin (RR) CPU Scheduling Algorithm. The program should accept the number of processes, their Arrival Times (AT), Burst Times (BT), and a Time Quantum (TQ). The CPU should rotate through the processes in the ready queue, granting each a maximum of one TQ of execution time. Calculate the Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) for each process and the overall averages.

Input Format

1. The first line of input contains an integer representing the number of processes.
2. The second line contains an integer representing the Time Quantum.
3. The next lines contain two integers for each process, where

- The first integer denotes the Arrival Time (AT)
- The second integer denotes the Burst Time (BT) of the process.

Output Format

The output displays the Average Turnaround Time and the Average Waiting Time.

Select Language : C (GCC 9.2.0)

```c
        // If no process executed, CPU idle
        if(executed == 0)
            current_time++;
    }

    // Calculate TAT and WT
    for(i = 0; i < n; i++) {
        TAT[i] = CT[i] - AT[i];
        WT[i] = TAT[i] - BT[i];

        totalTAT += TAT[i];
        totalWT += WT[i];
    }

    printf("\nAverage Turnaround Time: %.2f\n", totalTAT / n);
    printf("Average Waiting Time: %.2f\n", totalWT / n);

    return 0;
}
```

Test Cases 2

Run Sample Cases

Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

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Output Format

The output displays the Average Turnaround Time and the Average Waiting Time.

Select Language : C (GCC 9.2.0)

```c
#include <stdio.h>

int main() {
    int n, tq, i;

    printf("Enter number of processes: \n");
    scanf("%d", &n);

    printf("Enter Time Quantum: \n");
    scanf("%d", &tq);

    int AT[n], BT[n], RT[n], CT[n], TAT[n], WT[n];

    // Input
    for(i = 0; i < n; i++) {
        printf("Enter AT and BT for P%d: \n", i + 1);
        scanf("%d %d", &AT[i], &BT[i]);
        RT[i] = BT[i];  // Remaining Time
    }
```

Test Cases 2

Run Sample Cases

Run All Cases

Custom Test Case

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Sample Test Case

> Test Case - 1